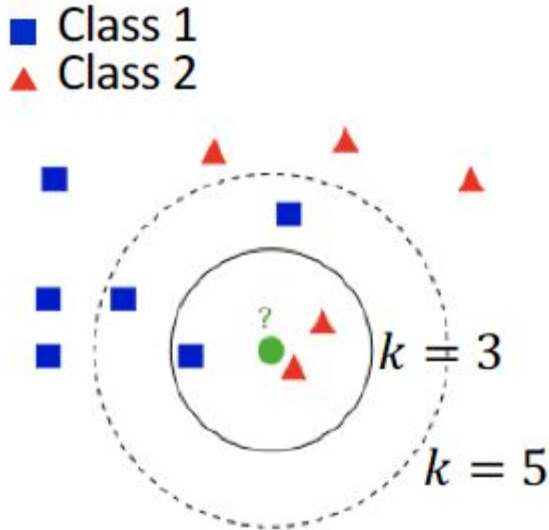


# K-Nearest Neighbors (KNN)

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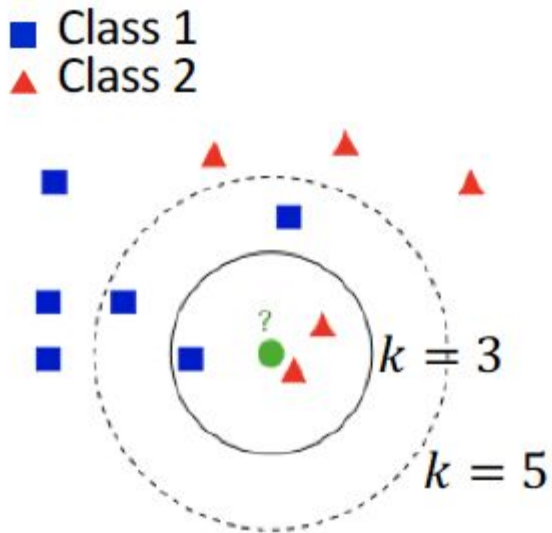
# KNN

- Simple, but powerful
- Make predictions on previously unseen data based on nearby data points from the training set

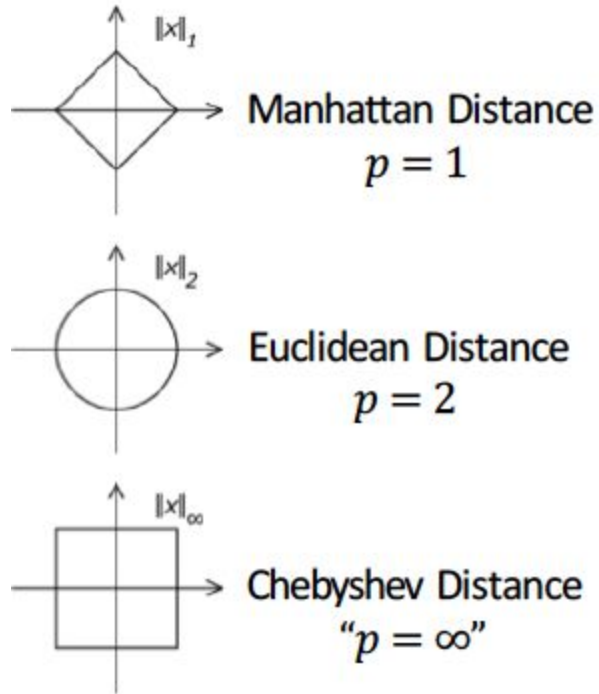


# KNN

- Majority Vote



# Distance Measure



# Euclidean Distance

$$d(\vec{x}, \vec{y})_2 = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

# Manhattan Distance

$$d(\vec{x}, \vec{y})_1 = \|\vec{x} - \vec{y}\|_1 = \sum_{i=1}^n |x_i - y_i|$$

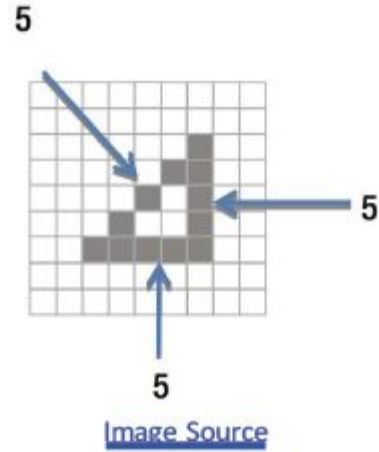


# Chebyshev Distance

Chebyshev distance, is

$$d_{\max}(\vec{x}, \vec{y}) = \lim_{p \rightarrow \infty} \left( \sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

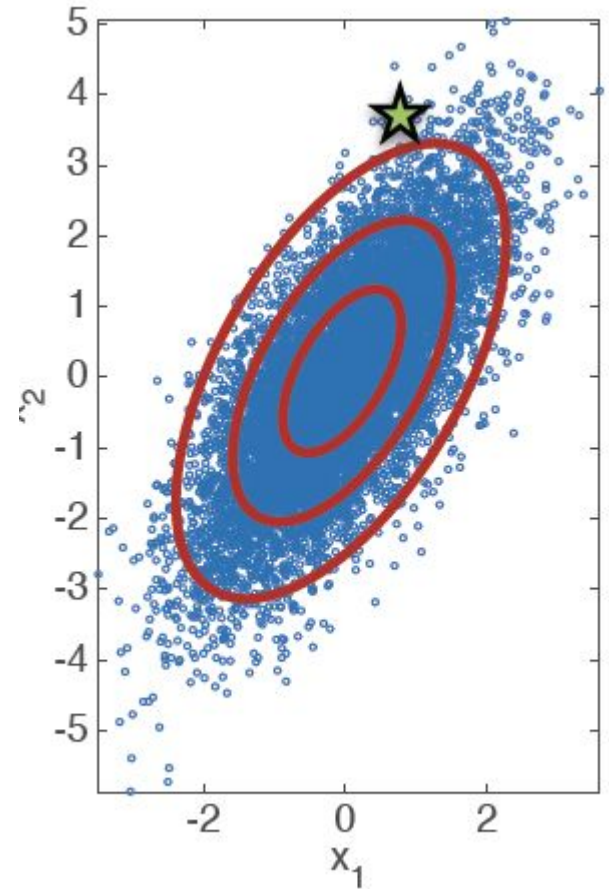
$$= \max_i |x_i - y_i|$$



|   | a | b | c | d | e | f                                                                                   | g | h |   |
|---|---|---|---|---|---|-------------------------------------------------------------------------------------|---|---|---|
| 8 | 5 | 4 | 3 | 2 | 2 | 2                                                                                   | 2 | 2 | 8 |
| 7 | 5 | 4 | 3 | 2 | 1 | 1                                                                                   | 1 | 2 | 7 |
| 6 | 5 | 4 | 3 | 2 | 1 |  | 1 | 2 | 6 |
| 5 | 5 | 4 | 3 | 2 | 1 | 1                                                                                   | 1 | 2 | 5 |
| 4 | 5 | 4 | 3 | 2 | 2 | 2                                                                                   | 2 | 2 | 4 |
| 3 | 5 | 4 | 3 | 3 | 3 | 3                                                                                   | 3 | 3 | 3 |
| 2 | 5 | 4 | 4 | 4 | 4 | 4                                                                                   | 4 | 4 | 2 |
| 1 | 5 | 5 | 5 | 5 | 5 | 5                                                                                   | 5 | 5 | 1 |
|   | a | b | c | d | e | f                                                                                   | g | h |   |

# Mahalanobis Distance

$$d_{\text{Mahalanobis}}(\vec{x}, \vec{\mu}) = \sqrt{(\vec{x} - \vec{\mu})^T \vec{S}^{-1} (\vec{x} - \vec{\mu})}$$







<https://www.youtube.com/watch?v=0p0o5cmgLdE>